

8 Building the body

We would be following the differential drive type locomotion technique as it is easy to implement and very efficient as well. Also, you don't need to go for wood to build a body for the line follower since you do not require your robot to have that much your strength. The best material to make the body for your first robot is the cardboard. Take any cardboard box which you may have at your home. It will already be in shape and you would be spared with the trouble of designing a body, fitting various parts of the structure together. A cardboard box of right size would be perfect for application. The right size would be according to the size of your motor. Ideally, it should be as wide as twice the length of your motor and its length should be such that it could accommodate the breadboard on its top. I found one such box and I believe it is just perfect for our application.



After selecting the material for body, we need to decide which motor we should use. The DC geared motors provide good torque, so we would be using them. Also, we don't need too much speed in locomotion. Too much speed would require a very fast processor and efficient code since if the speed is high, the processing unit will have lesser time to judge a turn

and respond. So, a DC geared motor with the speed of 50-100 rpm would be ideal for us.

The next step is to make holes on the sides such from where the motor's axle would come out. The diameter of this whole should be equal to the width of the axle of your motor. Following figure shows the circular area which is to be removed for making the hole.

Note that the height of the hole from the base would be equal to the height of the axle of the motor from

